Melanoma: Cutaneous, Version 2.2026

Susan M. Swetter, MD[1,*]; Douglas Johnson, MD, MSCI[2,*]; Mark R. Albertini, MD[3,*]; Christopher A. Barker, MD[4]; Sarah Bateni, MD, MAS[5]; Joel Baumgartner, MD[6]; Shailender Bhatia, MD[7]; Christopher Bichakjian, MD[8]; Genevieve Boland, MD, PhD[9]; Sunandana Chandra, MD, MS[10]; David Chen, MD, PhD[11]; Bartosz Chmielowski, MD, PhD[12]; Jacob Choi, MD, PhD[10]; Dominick DiMaio, MD[13]; Roxana Dronca, MD[14]; Martin D. Fleming, MD[15]; Anjela Galan, MD[16]; Samantha Guild[17]; Siwen Hu-Lieskovan, MD, PhD[18]; Giorgos Karakousis, MD[19]; Kari Kendra, MD, PhD[20]; Maija Kiuru, MD, PhD[21]; Julie R. Lange, MD, ScM[22]; Ryan Lanning, MD, PhD[23]; Theodore Logan, MD[24]; Miguel Materin, MD[25]; Kelly Nelson, MD[26,*]; Daniel Olson, MD[27]; Anthony J. Olszanski, MD, RPh[28]; Patrick A. Ott, MD, PhD[29]; Igor Puzanov, MD, MSCI[30]; Luke Rothermel, MD, MPH[31]; April K. Salama, MD[25]; Adrienne Shannon, MD[32]; Rohit Sharma, MD[32]; Arun Singh, MD[31]; Andrew Stacey, MD, MS[7]; Katy Tsai, MD[33]; Evan Wuthrick, MD[34]; Yan Xing, MD, PhD[35]; Nicole McMillian, MS[36,*]; and Sara Espinosa, PhD[36,*]

Abstract

The NCCN Guidelines for Melanoma: Cutaneous provide strategies for the diagnosis, treatment, and follow-up of patients with cutaneous melanoma, which is among the most common cancers in the United States. The outcome of cutaneous melanoma is significantly influenced by stage at initial presentation. Generally, localized disease and smaller primary tumors have been associated with more favorable prognosis. Thus, the select content featured in this issue focuses on risk factors associated with cutaneous melanoma development and which should be considered during an initial assessment. These include (1) phenotypic predispositions, (2) personal medical history or comorbidities, (3) genetic predispositions, and (4) environmental factors.

J Natl Compr Canc Netw 2026;24(6):272–277
doi:10.6004/jnccn.2026.0027

Overview

In 2026, an estimated 112,000 patients will be diagnosed with cutaneous melanoma and about 8,510 patients will die of disease in the United States.[1] Although melanoma incidence has steadily increased over the past several decades, this has largely been due to the diagnoses of melanomas in situ and early stage (T1) melanomas, raising concerns for potential histopathologic overinterpretation.[2] However, data have shown that the thickest (T4) melanomas continue to rise in incidence, particularly in individuals with lower socioeconomic status and/or skin of color.[2] Risk factors for melanoma include individuals assigned male at birth (AMAB), older age (typically >50 years), phenotypic and genetic predisposition, and personal medical history, comorbidities, and environmental factors (see "Risk Factors for Development of Single or Multiple Primary Melanomas," Figure 1).[1,3–27] Melanoma occurs most often in non-Hispanic White individuals with lighter skin tones but can occur in people with darker skin tones from any racial or ethnic group, and in areas of the body without substantial sun exposure (eg, palms, soles, nails).

As with nearly all malignancies, the outcome of cutaneous melanoma depends on the stage at presentation.[28] In general, the prognosis is excellent for patients who present with localized disease and primary tumors ≤1.0 mm in thickness, with 5-and 10-year disease-specific survival (DSS) achieved in >96% of patients.[28] For patients with localized melanomas >1.0 mm in thickness, 5- and 10-year DSS range from 75% to 94%, depending on tumor thickness, ulceration, and mitotic rate.[28] When regional nodes are involved, 5-and 10-year DSS range between 24% and 93%, depending on the extent of nodal involvement, although these survival rates exclude data following the use of novel targeted and immune therapies. Patients with clinically occult nodal disease (detected via sentinel lymph node biopsy) generally have better survival than those with clinically evident nodal disease.[28–30] Historically, long-term survival in patients with distant metastatic melanoma was <10% at 5 years, with median survival <1 year.[31–34] However, beginning in the early 2010s, effective systemic therapies for patients with unresectable stage III and stage IV melanoma made long-term remission possible for approximately 50% of patients, and these agents are increasingly being used in the adjuvant setting to prevent disease relapse and improve survival.[35]

[1]Stanford Cancer Institute; [2]Vanderbilt-Ingram Cancer Center; [3]University of Wisconsin Carbone Cancer Center; [4]Memorial Sloan Kettering Cancer Center; [5]O'Neal Comprehensive Cancer Center at UAB; [6]UC San Diego Moores Cancer Center; [7]Fred Hutchinson Cancer Center; [8]University of Michigan Rogel Cancer Center; [9]Mass General Cancer Center; [10]Robert H. Lurie Comprehensive Cancer Center of Northwestern University; [11]Siteman Cancer Center at Barnes-Jewish Hospital and Washington University School of Medicine; [12]UCLA Jonsson Comprehensive Cancer Center; [13]Fred & Pamela Buffett Cancer Center; [14]Mayo Clinic Comprehensive Cancer Center; [15]The University of Tennessee Health Science Center; [16]Yale Cancer Center/Smilow Cancer Hospital; [17]AIM at Melanoma; [18]Huntsman Cancer Institute at the University of Utah; [19]Abramson Cancer Center at the University of Pennsylvania; [20]The Ohio State University Comprehensive Cancer Center - James Cancer Hospital and Solove Research Institute; [21]UC Davis Comprehensive Cancer Center; [22]Johns Hopkins Kimmel Cancer Center; [23]University of Colorado Cancer Center; [24]Indiana University Melvin and Bren Simon Comprehensive Cancer Center; [25]Duke Cancer Institute; [26]The University of Texas MD Anderson Cancer Center; [27]The UChicago Medicine Comprehensive Cancer Center; [28]Fox Chase Cancer Center; [29]Dana-Farber/Brigham and Women's Cancer Center; [30]Roswell Park Comprehensive Cancer Center; [31]Case Comprehensive Cancer Center/University Hospitals Seidman Cancer Center and Cleveland Clinic Taussig Cancer Institute; [32]UT Southwestern Simmons Comprehensive Cancer Center; [33]UCSF Helen Diller Family Comprehensive Cancer Center; [34]Moffitt Cancer Center; [35]City of Hope National Medical Center; and [36]National Comprehensive Cancer Network.

*Discussion Writing Committee Member.

To view disclosures of external relationships for the NCCN Guidelines panel, go to https://www.nccn.org/guidelines/guidelines-panels-and-disclosure/disclosure-panels

The full NCCN Guidelines for Melanoma: Cutaneous are not printed in this issue of *JNCCN*. The complete and most recent version of these guidelines is available free of charge at NCCN.org.

NCCN CATEGORIES OF EVIDENCE AND CONSENSUS

Category 1: Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A: Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B: Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3: Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

NCCN CATEGORIES OF PREFERENCE

Preferred intervention: Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other: Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain: Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

PLEASE NOTE

The NCCN Guidelines® are a statement of evidence and consensus of the authors regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult the NCCN Guidelines® is expected to use independent medical judgment in the context of individual clinical circumstances to determine any patient's care or treatment. The National Comprehensive Cancer Network® (NCCN®) makes no representations or warranties of any kind regarding their content, use, or application and disclaims any responsibility for their application or use in any way.

The currently described clinical subtypes of cutaneous melanoma are low cumulative sun damage (CSD; low-CSD, typically superficial spreading or nodular subtypes), high CSD (typically lentigo maligna subtype, signified by the presence of marked solar elastosis and lentigo maligna growth pattern histologically), and acral melanoma on the soles, palms, or subungual sites. Melanocytes outside the skin can give rise to noncutaneous melanomas on mucosal membranes (eg, in the head and neck sinuses and oral cavity, anorectum, vulva, vagina, and gastrointestinal and urogenital tracts) and the uveal tract of the eye, or leptomeninges.[36,37] Mucosal and uveal melanomas differ significantly from cutaneous melanoma in presentation, genetic profile, staging, response to treatment, and patterns of progression, and should be treated as diseases distinct from cutaneous melanoma (see NCCN Guidelines for Head and Neck Cancers, NCCN Guidelines for Melanoma: Uveal, and NCCN Guidelines for Vulvar Cancer, available at NCCN.org).[38–41]

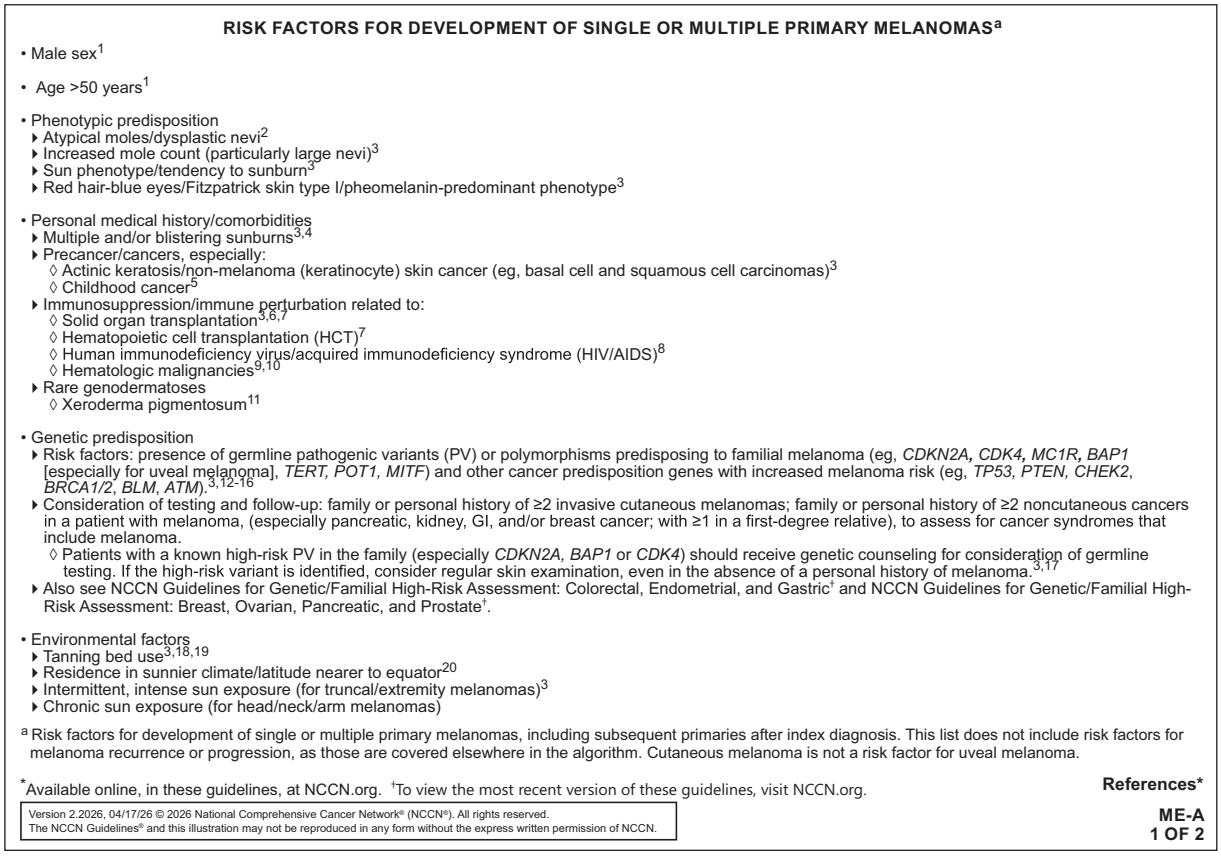


Figure 1. ME-A. NCCN Clinical Practice Guidelines in Oncology for Melanoma: Cutaneous, Version 2.2026.

Risk Factors for Development of Single or Multiple Primary Melanomas

During the initial presentation of a suspicious pigmented skin lesion, patients should be assessed for melanoma-related risk factors (see Figure 1). These risk factors include individuals assigned male at birth and age >50 years[1]; additional factors include phenotypic predispositions, personal medical history/comorbidities, genetic predispositions, and environmental factors, particularly, ultraviolet radiation (ultraviolet-R) exposure. This list does not include risk factors for melanoma recurrence or progression. Additionally, cutaneous melanoma is not a risk factor for uveal or mucosal melanoma. However, studies have shown that patients with Parkinson disease and survivors of non-Hodgkin lymphoma or chronic lymphocytic leukemia/)/small lymphocytic lymphoma are also at increased risk of developing cutaneous melanomas.[8,9,42,43]

Phenotypic Predispositions

Melanocytic nevi of any type (common/typical, atypical/dysplastic, or congenital) can serve as precursors to melanoma, though the vast majority (>80%–90%) of melanomas are believed to arise de novo. Common/typical nevi are usually smaller in diameter (≤5 mm) and have more uniform coloring than clinically atypical/dysplastic nevi, which share some clinical features with melanoma (larger size, color variation), but are deemed benign.[3,44] Atypical/dysplastic nevi serve as risk markers for melanoma developing elsewhere in the body, especially in the setting of higher overall nevus count.[3,6] Melanomas arising from a nevus tend to have a more favorable histology than melanomas that are not associated with a nevus.[45] Significant interobserver histopathologic variability in the labeling of melanocytic neoplasms (ranging from moderately dysplastic to thin [T1a] melanoma) has contributed to overinterpretation (or "overcalling") of melanoma and led to reclassification in the M-PATH schema as well as further investigation of pathology, relabeling melanomas in situ and severely dysplastic nevi as lower risk neoplasms.[46,47]

Other risk factors for melanoma include difficulty tanning, light skin tone/tendency to sunburn, red hair, blue eyes, and a high tendency to develop freckles.[48,49] Skin pigmentation is determined by the ratio between eumelanin (brown-black pigment) and pheomelanin (yellow-red pigment).[50] A pheomelanin-predominant phenotype is generally indicative of lighter skin color, lower Fitzpatrick skin type, and an increased sensitivity to sunlight with which people are more likely to burn than tan.[6,50]

Personal Medical History or Comorbidities

Other risk factors for the development of melanoma related to personal medical history or comorbidities include multiple blistering sunburns, precancers/cancers including keratinocyte (basal cell and squamous cell) carcinomas and childhood cancer, immunosuppression such as that related to HIV/AIDS, solid organ transplantation, hematopoietic cell transplantation (HCT), and rare genodermatoses like xeroderma pigmentosum.

In a large prospective study of White females in the United States (n=108,916) with a 20-year follow-up, it was determined that certain subtypes of melanoma (superficial spreading and nodular types) were primarily associated with sun exposure during early life instead of adulthood.[7] This determination was reached by defining ≥5 blistering sunburns between age 15 and 20 years as an indicator for early life sun exposure.[7] Additionally, sun exposure meta-analysis supports that sunburn history plays a role as a melanoma risk factor.[6,23] Other melanoma subtypes that tend to occur on sun-exposed skin (eg, lentigo maligna and desmoplastic types) are associated with chronic sun exposure and occur later in life. Acral and mucosal melanomas are not related to ultraviolet radiation exposure.

In a study investigating the risk of developing melanoma in patients with a history of basal cell carcinoma (BCC) and/or squamous cell carcinomas (SCCs), 10 of the 290 patients developed malignant melanoma within a follow-up average of 109 months. All melanomas were <0.70 mm in Breslow thickness and most occurred on usually clothed cutaneous sites. The reported relative risk (RR) was 17 when compared with a control population indicating that patients with BCC/SCC skin cancer are at significant increased risk for developing melanomas, related to shared etiologic risk through ultraviolet-R exposure. Surveillance through annual skin examinations is an excellent method for early detection of curable melanoma lesions in these patients.[51] Other studies have also indicated that a medical history of non-melanoma skin cancers is associated with the development of other primary malignancies and increased cancer mortality.[52,53]

Having a history of childhood cancer is also a risk factor for subsequent malignancy. In a study of 14,358 5-year survivors of childhood cancer diagnosed between 1970 and 1986, 57 melanomas (46 invasive, 2 ocular, and 9 in situ) occurred in 51 survivors.[10] The median time and patient age for the development of melanoma was 21.0 and 32.3 years, respectively. Initial childhood cancer diagnoses predominantly included soft tissue and bone sarcoma, leukemia, lymphoma, and central nervous system malignancy.[10] In this study, family history of cancer, demographic, or treatment-related factors had no statistically significant associations with melanoma risk. However, there was a 2.5-fold increased melanoma risk in childhood cancer survivors, indicating that early screening and prevention are essential.

Melanoma is an immune-responsive tumor and so can occur at higher rates and respond differently to treatment in immunocompromised populations. Individuals with HIV/AIDS are reported to be at significantly increased risk for developing melanoma.[13] One study indicated that individuals with lighter skin with HIV/AIDS should receive regular skin screening and counseled against excessive sun exposure. Because these hematologic malignancies confer increased melanoma risk associated with their relative immunosuppression, see the NCCN Guidelines for Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma (available at NCCN.org).

In a comparative study, melanoma incidence, including risk of regional involvement, was evaluated among 139,991 non-Hispanic White transplant recipients using linked US transplant-cancer registry data (1987–2010).[11] Receipt of certain immunosuppressive therapies, such as azathioprine maintenance or polyclonal antibody induction, correlated with increased risk. Organ transplant recipients are also reported to have a higher melanoma-specific mortality than nonrecipients because they remain on immunosuppression long term to prevent graft rejection.[11,54] Additionally, a 2024 study evaluating 901 non-Hispanic White organ transplant recipients with melanoma diagnosed before transplantation determined that this patient population was at a 9-fold increased risk for recurrence and 52-fold increase for melanoma-specific mortality after transplantation.[55]

There is also a reported risk of skin cancer in HCT recipients when compared with individuals who have not received any transplant. A nationwide population-based Danish study included 3,302 patients who underwent allogeneic or autologous HCT, 4,789 renal transplant recipients, and 10 age- and sex-matched nontransplanted individuals for each population group. The study determined that, in comparison with the background population, allogeneic HCT recipients had an increased risk of BCC, SCC, or melanoma. Additionally, when compared with recipients of renal transplants, recipients of allogeneic HCT had a 3-fold higher risk of melanoma. Conversely, no increased risk of skin cancer was observed in recipients of autologous HCT.[12]

Xeroderma pigmentosum (XP) is a rare genodermatosis (genetic skin disorder with multisystem involvement) that prevents DNA repair from sun damage, leading to an ultraviolet light sensitivity in about 50% of patients.[56] Patients with XP should practice sun avoidance and protection to prevent premalignant or malignant lesions. Kraemer et al estimated the clinical frequency of skin cancer from a medical literature survey of 830 patients with XP, determining that the median patient age of skin cancer development was 12 years with no gender preference while the average age of symptom onset was between 1 and 2 years.[14] Melanoma was reported in 5% of these patients and within this population, skin cancer most frequently presented on the face, head, or neck, which are areas of skin that receive more frequent sun exposure.

Genetic Predispositions

Generally, a family history of melanoma appears to increase risk of melanoma.[57] Previous cutaneous melanomas place an individual at high-risk of developing additional primary melanomas.[58,59] About 5% of patients who develop melanoma without a family or prior personal history of melanoma develop subsequent new primary cutaneous melanoma(s), whereas in those with a family history of melanoma, the corresponding estimate is about 10%.[58–60] These reports indicate that the risk of a second primary melanoma is greater for males and older patients. A Swedish study including >65,000 individuals with melanoma determined a standardized incidence ratio (SIR) of 2.8 for a second melanoma in individuals with a family history of melanoma and a SIR of 2.5 in individuals with no family history.[61] The risk of a second melanoma also increased when the first melanoma was diagnosed at <40 years.

Approximately 10% of melanomas can be linked to hereditary factors, including the presence of germline pathogenic variants (PV) or polymorphisms predisposing to familial melanomas. In a recent, prospectively accrued cohort of 400 individuals with melanoma and personal or family history of cancers who underwent germline testing of >80 cancer predisposition genes, pathogenic or likely pathogenic variants were identified in 61 (15.3%). Three validation cohorts demonstrated pathogenic or likely pathogenic rates between 10.6% and 15.8%.[62] Germline PVs or polymorphisms associated with melanoma predisposition include *CDKN2A*, *CDK4*, *MC1R*, *BAP1* (especially for uveal melanoma), *TERT*, *POT1*, and *MITF*.[6,15–17,62,63] Other cancer predisposition genes with increased melanoma risk include *TP53*, *PTEN*, *CHEK2*, *BRCA1*, *BRCA2*, *BLM*, and *ATM*. The NCCN Melanoma Panel recommends referral to a cancer genetics specialist for consideration of testing and follow-up for individuals with a family or personal history of ≥2 invasive cutaneous melanomas or those with a family or personal history of ≥2 noncutaneous cancers in a patient with melanoma, especially renal, gastrointestinal, and/or breast cancer in at least one first-degree relative, to assess for cancer syndromes that include melanoma. The panel also recommends genetic counseling for consideration of germline testing even if the patient has no history of melanoma in 2 scenarios: (1) if there is a known high-risk PV (*CDKN2A*, *BAP1*, or *CDK4*) in the family; or (2) if the patient or a first-degree relative has a history of pancreatic cancer. Regular skin examination should be considered if a high-risk PV is identified.[6,18] For additional information about genetic testing, also see the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric and NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate (available at NCCN.org).

Environmental Factors

Environmental factors, namely excessive ultraviolet-R exposure, whether natural (sunlight) or artificial (tanning beds) are arguably the most important preventable/modifiable risk factors for melanoma development in people with lighter skin tones or a tendency to sunburn. Ultraviolet-R exposure is believed to be related to approximately 90% of melanomas in White individuals, and tanning bed use has significantly increased melanoma risk and incidence worldwide.[64,65]

Tanning or sunbed use has been associated with increased risk of melanoma incidence in various studies.[6,19,20] A population-based case-controlled study included 1,167 patients with melanoma and 1,101 control subjects between 25 and 29 years of age.[19] Approximately 63% of patients with melanoma and 51% of the control population reported tanning indoors. Melanoma risk was more prominent in use of UVA versus UVB devices, and the risk increased with the additional time using these devices; the number of years during which tanning beds were used was especially relevant for the development of melanoma lesions. Although this study found that onset of tanning bed use did not impact increased risk, Cust et al,[20] whose patient population was 18 to 29 years of age, found that an earlier age at first use was associated with earlier onset of disease. A recent case/control analysis of more than 2,932 dermatology patients with and 2,925 without history of tanning bed use showed a 2.85-fold higher risk of melanoma in users, after adjusting for age, sex, family history of melanoma, and sunburn history.[66] A dose-response relationship was observed, with the RR for melanoma significantly increased by the number of tanning bed exposures. Additional molecular analyses performed in the study showed a nearly 2-fold increased mutation burden of melanocytes in tanning bed users compared with nonusers that could serve as drivers of melanoma.

Studies worldwide have confirmed the strong link of ultraviolet-R exposure and cutaneous melanoma, particularly in individuals with lighter skin tones who live in warmer climates or closer to the equator.[67] To evaluate the risk of melanoma in patients with intermittent sun exposure and sunburn history, a meta-analysis extracted RR for sun exposure from 57 studies and discovered a correlation to higher risk.[6] Additionally, variability of the intermittent sun exposure estimates were affected by the country where the study took place.[6]

Various occupations where chemical exposure is more likely may also present an elevated risk for melanoma. For example, studies suggest that firefighters are at higher risk due to increased exposure to carcinogens in retardants and by-products of combustion compared with the general population, in addition to

insufficient skin cancer screening practices.[68–71] Male firefighters in Florida had an adjusted odds ratio of 1.56 for melanoma with a stronger association in male firefighters diagnosed <50 years (adjusted odds ratio, 1.87), whereas the risk for melanoma for female firefighters approached but did not reach statistical significance in this study.[72] In a large cohort of Danish male firefighters (n=11,827) the SIR for melanoma was 1.30 and 1.37 in individuals who had been employed as a firefighter for >5 years.[73] A cross-sectional study of 552 volunteer firefighters in the United States noted that only 26.1% received skin cancer screening, especially among individuals who did not use sunscreen.[71] Although results from meta-analyses support a causal relationship between a firefighter's occupational exposures and higher cancer risk, additional longitudinal studies with exposure-specific parameters are needed to further explore this link and develop appropriate screening recommendations.[69,70]

Melanoma is the fourth most common cancer among US veterans,[74] and deployment-related hazard of prolonged ultraviolet-R exposure in sunny regions contributes to skin cancer risk.[75–78] Melanoma is also on the list of presumptive cancers for eligible Gulf War and post-9/11 veterans who have melanoma and were exposed to burn pits or served in a qualifying location.[79] Veterans are not only at higher risk for melanoma (including acral melanoma due to Agent Orange exposure), but they are also more likely to present with advanced-stage disease.[80]

Summary

The NCCN Guidelines for Melanoma: Cutaneous provide an evidence- and consensus-based framework for the management of cutaneous melanoma across the entire clinical spectrum. As early detection of cutaneous melanomas is key to favorable outcomes, the melanoma panel has defined risk factors that can influence the development of cutaneous melanomas. These risk factors should be considered by physicians at the initial clinical presentation.

References

1. Siegel RL, Kratzer TB, Wagle NS, et al. Cancer statistics, 2026. CA Cancer J Clin 2026;76:e70043.
2. Chen ML, de Vere Hunt IJ, John EM, et al. Differences in thickness-specific incidence and factors associated with cutaneous melanoma in the US from 2010 to 2018. JAMA Oncol 2022;8:755–759.
3. Rigel DS, Rivers JK, Kopf AW, et al. Dysplastic nevi. Markers for increased risk for melanoma. Cancer 1989;63:386–389.
4. Colantonio S, Bracken MB, Beecker J. The association of indoor tanning and melanoma in adults: systematic review and meta-analysis. J Am Acad Dermatol 2014;70:847–857.e1–18.
5. Gordon D, Gillgren P, Eloranta S, et al. Time trends in incidence of cutaneous melanoma by detailed anatomical location and patterns of ultraviolet radiation exposure: a retrospective population-based study. Melanoma Res 2015;25:348–356.
6. Chen ST, Geller AC, Tsao H. Update on the epidemiology of melanoma. Curr Dermatol Rep 2013;2:24–34.
7. Wu S, Han J, Laden F, Qureshi AA. Long-term ultraviolet flux, other potential risk factors, and skin cancer risk: a cohort study. Cancer Epidemiol Biomarkers Prev 2014;23:1080–1089.
8. Lam CJ, Curtis RE, Dores GM, et al. Risk factors for melanoma among survivors of non-hodgkin lymphoma. J Clin Oncol 2015;33:3096–3104.
9. Olsen CM, Lane SW, Green AC. Increased risk of melanoma in patients with chronic lymphocytic leukaemia: systematic review and meta-analysis of cohort studies. Melanoma Res 2016;26:188–194.
10. Pappo AS, Armstrong GT, Liu W, et al. Melanoma as a subsequent neoplasm in adult survivors of childhood cancer: a report from the childhood cancer survivor study. Pediatr Blood Cancer 2013;60:461–466.
11. Robbins HA, Clarke CA, Arron ST, et al. Melanoma risk and survival among organ transplant recipients. J Invest Dermatol 2015;135:2657–2665.
12. Omland SH, Gniadecki R, Hædersdal M, et al. Skin cancer risk in hematopoietic stem-cell transplant recipients compared with background population and renal transplant recipients: a population-based cohort study. JAMA Dermatol 2016;152:177–183.
13. Olsen CM, Knight LL, Green AC. Risk of melanoma in people with HIV/AIDS in the pre- and post-HAART eras: a systematic review and meta-analysis of cohort studies. PLoS One 2014;9:e95096.
14. Kraemer KH, Lee MM, Scotto J. Xeroderma pigmentosum. Cutaneous, ocular, and neurologic abnormalities in 830 published cases. Arch Dermatol 1987;123:241–250.
15. Bubien V, Bonnet F, Brouste V, et al. High cumulative risks of cancer in patients with PTEN hamartoma tumour syndrome. J Med Genet 2013;50:255–263.
16. Tan MH, Mester JL, Ngeow J, et al. Lifetime cancer risks in individuals with germline PTEN mutations. Clin Cancer Res 2012;18:400–407.
17. Leachman SA, Lucero OM, Sampson JE, et al. Identification, genetic testing, and management of hereditary melanoma. Cancer Metastasis Rev 2017;36:77–90.
18. Chen T, Hemminki K, Kharazmi E, et al. Multiple primary (even in situ) melanomas in a patient pose significant risk to family members. Eur J Cancer 2014;50:2659–2667.
19. Lazovich D, Vogel RI, Berwick M, et al. Indoor tanning and risk of melanoma: a case-control study in a highly exposed population. Cancer Epidemiol Biomarkers Prev 2010;19:1557–1568.
20. Cust AE, Armstrong BK, Goumas C, et al. Sunbed use during adolescence and early adulthood is associated with increased risk of early-onset melanoma. Int J Cancer 2011;128:2425–2435.
21. Richards TB, Johnson CJ, Tatalovich Z, et al. Association between cutaneous melanoma incidence rates among white US residents and county-level estimates of solar ultraviolet exposure. J Am Acad Dermatol 2011;65:S50–57.
22. Caini S, Gandini S, Sera F, et al. Meta-analysis of risk factors for cutaneous melanoma according to anatomical site and clinico-pathological variant. Eur J Cancer 2009;45:3054–3063.
23. Gandini S, Sera F, Cattaruzza MS, et al. Meta-analysis of risk factors for cutaneous melanoma: II. Sun exposure. Eur J Cancer 2005;41:45–60.
24. Whiteman DC, Stickley M, Watt P, et al. Anatomic site, sun exposure, and risk of cutaneous melanoma. J Clin Oncol 2006;24:3172–3177.
25. Ghiasvand R, Robsahm TE, Green AC, et al. Association of phenotypic characteristics and UV radiation exposure with risk of melanoma on different body sites. JAMA Dermatol 2019;155:39–49.
26. Arnold M, de Vries E, Whiteman DC, et al. Global burden of cutaneous melanoma attributable to ultraviolet radiation in 2012. Int J Cancer 2018;143:1305–1314.
27. Berge LAM, Andreassen BK, Stenehjem JS, et al. Use of immunomodulating drugs and risk of cutaneous melanoma: a nationwide nested case-control study. Clin Epidemiol 2020;12:1389–1401.
28. Gershenwald JE, Scolyer RA, Hess KR, et al. Melanoma staging: evidence-based changes in the American Joint Committee on Cancer eighth edition cancer staging manual. CA Cancer J Clin 2017;67:472–492.
29. Balch CM, Gershenwald JE, Soong SJ, et al. Multivariate analysis of prognostic factors among 2,313 patients with stage III melanoma: comparison of nodal micrometastases versus macrometastases. J Clin Oncol 2010;28:2452–2459.
30. Cascinelli N, Belli F, Santinami M, et al. Sentinel lymph node biopsy in cutaneous melanoma: the WHO Melanoma Program experience. Ann Surg Oncol 2000;7:469–474.
31. Manola J, Atkins M, Ibrahim J, Kirkwood J. Prognostic factors in metastatic melanoma: a pooled analysis of eastern cooperative oncology group trials. J Clin Oncol 2000;18:3782–3793.

32. Barth A, Wanek LA, Morton DL. Prognostic factors in 1,521 melanoma patients with distant metastases. J Am Coll Surg 1995;181:193–201.
33. Balch CM. Cutaneous melanoma: prognosis and treatment results worldwide. Semin Surg Oncol 1992;8:400–414.
34. Unger JM, Flaherty LE, Liu PY, et al. Gender and other survival predictors in patients with metastatic melanoma on Southwest Oncology Group trials. Cancer 2001;91:1148–1155.
35. Singh SRK, Malapati SJ, Kumar R, et al. NCDB analysis of melanoma 2004-2015: epidemiology and outcomes by subtype, sociodemographic factors impacting clinical presentation, and real-world survival benefit of immunotherapy approval. Cancers (Basel) 2021;13:1455.
36. Kibbi N, Kluger H, Choi JN. Melanoma: clinical presentations. Cancer Treat Res 2016;167:107–129.
37. Mihajlovic M, Vlajkovic S, Jovanovic P, Stefanovic V. Primary mucosal melanomas: a comprehensive review. Int J Clin Exp Pathol 2012;5:739–753.
38. Curtin JA, Busam K, Pinkel D, Bastian BC. Somatic activation of KIT in distinct subtypes of melanoma. J Clin Oncol 2006;24:4340–4346.
39. Shields CL, Kaliki S, Furuta M, et al. American joint committee on cancer classification of uveal melanoma (anatomic stage) predicts prognosis in 7,731 patients: the 2013 Zimmerman lecture. Ophthalmology 2015;122:1180–1186.
40. Tacastacas JD, Bray J, Cohen YK, et al. Update on primary mucosal melanoma. J Am Acad Dermatol 2014;71:366–375.
41. Singh M, Durairaj P, Yeung J. Uveal melanoma: a review of the literature. Oncol Ther 2018;6:87–104.
42. Bertoni JM, Arlette JP, Fernandez HH, et al. Increased melanoma risk in Parkinson disease: a prospective clinicopathological study. Arch Neurol 2010;67:347–352.
43. Flynn MS, Robinson C, Patel S, et al. Clinicopathologic characteristics of melanoma in patients with Parkinson disease. JID Innov 2023;3:100173.
44. Barnhill RL, Elder DE, Piepkorn MW, et al. Revision of the melanocytic pathology assessment tool and hierarchy for diagnosis classification schema for melanocytic lesions: a consensus statement. JAMA Netw Open 2023;6:e2250613.
45. Olsen CM, Carroll HJ, Whiteman DC. Estimating the attributable fraction for cancer: a meta-analysis of nevi and melanoma. Cancer Prev Res (Phila) 2010;3:233–245.
46. Wu Z, Nickel B, Boroumand F, et al. Impact of alternative diagnostic labels for melanoma in situ on management choices and psychological outcomes: protocol for an online randomised study. BMJ Open 2024;14:e089558.
47. Barnhill RL, Piepkorn MW, Duncan LM, et al. MPATH-Dx version 2.0 schema for melanocytic lesions: a robust tool for standardized diagnostic reporting. Clin Dermatol 2025;43:306–314.
48. Mitra D, Luo X, Morgan A, et al. An ultraviolet-radiation-independent pathway to melanoma carcinogenesis in the red hair/fair skin background. Nature 2012;491:449–453.
49. Rhodes AR, Weinstock MA, Fitzpatrick TB, et al. Risk factors for cutaneous melanoma. A practical method of recognizing predisposed individuals. JAMA 1987;258:3146–3154.
50. Schlessinger DI, Rahimi N, Schlessinger J. Biochemistry, Melanin. Treasure Island, FL: StatPearls Publishing; 2025.
51. Marghoob AA, Slade J, Salopek TG, et al. Basal cell and squamous cell carcinomas are important risk factors for cutaneous malignant melanoma. Screening implications. Cancer 1995;75:707–714.
52. Kahn HS, Tatham LM, Patel AV, et al. Increased cancer mortality following a history of nonmelanoma skin cancer. JAMA 1998;280:910–912.
53. Wheless L, Black J, Alberg AJ. Nonmelanoma skin cancer and the risk of second primary cancers: a systematic review. Cancer Epidemiol Biomarkers Prev 2010;19:1686–1695.
54. Engels EA, Pfeiffer RM, Fraumeni JF, et al. Spectrum of cancer risk among US solid organ transplant recipients. JAMA 2011;306:1891–1901.
55. Zwald FO, Sargen MR, Austin AA, et al. Outcomes in solid organ transplant recipients with a pretransplant diagnosis of melanoma. Am J Transplant 2024;24:993–1002.
56. Leung AK, Barankin B, Lam JM, et al. Xeroderma pigmentosum: an updated review. Drugs Context 2022;11:2022.
57. Wei EX, Li X, Nan H. Having a first-degree relative with melanoma increases lifetime risk of melanoma, squamous cell carcinoma, and basal cell carcinoma. J Am Acad Dermatol 2019;81:489–499.
58. Goggins WB, Tsao H. A population-based analysis of risk factors for a second primary cutaneous melanoma among melanoma survivors. Cancer 2003;97:639–643.
59. Slingluff CL, Vollmer RT, Seigler HF. Multiple primary melanoma: incidence and risk factors in 283 patients. Surgery 1993;113:330–339.
60. Soura E, Eliades PJ, Shannon K, et al. Hereditary melanoma: update on syndromes and management: genetics of familial atypical multiple mole melanoma syndrome. J Am Acad Dermatol 2016;74:395–407;quiz 408–310.
61. Chen T, Fallah M, Forsti A, et al. Risk of next melanoma in patients with familial and sporadic melanoma by number of previous melanomas. JAMA Dermatol 2015;151:607–615.
62. Funchain P, Ni Y, Heald B, et al. Germline cancer susceptibility in individuals with melanoma. J Am Acad Dermatol 2024;91:265–272.
63. Lochrin SE, Kemel Y, Stuart J, et al. Germline pathogenic variants in a large convenience cohort of multiple melanoma subtypes. J Clin Oncol 2024;42(Suppl):Abstract 9595.
64. Islami F, Sauer AG, Miller KD, et al. Cutaneous melanomas attributable to ultraviolet radiation exposure by state. Int J Cancer 2020;147:1385–1390.
65. International Agency for Research on Cancer Working Group on artificial ultraviolet (UV) light and skin cancer. The association of use of sunbeds with cutaneous malignant melanoma and other skin cancers: a systematic review. Int J Cancer 2007;120:1116–1122.
66. Gerami P, Tandukar B, Deivendran D, et al. Molecular effects of indoor tanning. Sci Adv 2025;11:eady4878.
67. Eide MJ, Weinstock MA. Association of UV index, latitude, and melanoma incidence in nonwhite populations–US Surveillance, Epidemiology, and End Results (SEER) Program, 1992 to 2001. Arch Dermatol 2005;141:477–481.
68. Teras LR, Diver WR, Mitchell EL, et al. Occupation as a firefighter and cancer mortality in a population-based cohort in the United States. Int J Epidemiol 2025;54:dyaf104.
69. Lee DJ, Ahn S, McClure LA, et al. Cancer risk and mortality among firefighters: a meta-analytic review. Front Oncol 2023;13:1130754.
70. DeBono NL, Daniels RD, Beane Freeman LE, et al. Firefighting and cancer: a meta-analysis of cohort studies in the context of cancer hazard identification. Saf Health Work 2023;14:141–152.
71. Shah NN, Steinberg MB, Caban-Martinez AJ, et al. Prevalence and predictors of skin cancer screening among a sample of US volunteer firefighters. Am J Ind Med 2023;66:897–903.
72. Lee DJ, Koru-Sengul T, Hernandez MN, et al. Cancer risk among career male and female Florida firefighters: evidence from the Florida Firefighter Cancer Registry (1981-2014). Am J Ind Med 2020;63:285–299.
73. Pedersen JE, Petersen KU, Andersen MG, et al. Cancer incidence in a cohort of Danish firefighters: an extended long-term follow-up 1968-2021. Am J Ind Med 2024;67:857–864.
74. Zullig LL, Sims KJ, McNeil R, et al. Cancer incidence among patients of the U.S. veterans affairs health care system: 2010 update. Mil Med 2017;182:e1883–1891.
75. Chang MS, La J, Trepanowski N, et al. Increased relative proportions of advanced melanoma among veterans: a comparative analysis with the surveillance, epidemiology, and end results registry. J Am Acad Dermatol 2022;87:72–79.
76. Powers JG, Patel NA, Powers EM, et al. Skin cancer risk factors and preventative behaviors among United States military veterans deployed to Iraq and Afghanistan. J Invest Dermatol 2015;135:2871–2873.
77. Rezaei SJ, Kim J, Onyeka S, et al. Skin cancer and other dermatologic conditions among US veterans. JAMA Dermatol 2024;160:1107–1111.
78. Riemenschneider K, Liu J, Powers JG. Skin cancer in the military: a systematic review of melanoma and nonmelanoma skin cancer incidence, prevention, and screening among active duty and veteran personnel. J Am Acad Dermatol 2018;78:1185–1192.
79. Library of Congress. H.R.3967 - 117th Congress (2021-2022): Honoring our PACT Act of 2022. Accessed April 1, 2026. Available at: https://www.congress.gov/bill/117th-congress/house-bill/3967
80. Hwang JC, Huhmann LB, Cho K, et al. Identification of risk factors for acral melanoma in US veterans. JAMA Dermatol 2026;162:279–291.